BAAZ & UDAY JKBOSE

2026 CLASS 11TH

SOME BASIC CONCEPT OF CHEMISTRY

LECTURE-08 CHEMISTRY

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Topics To Covered

1. STOICHIOMETRY AND STOICHIOMETRIC CALCULATION
2. LIMITING REAGENT
3. CONCENTRATION TERMS

Not mentioned in Syllabus.

(Given in NCERT).

REVISION.

EMPIRICAL FORMULA

- Ratio of different type of atoms present in molecule.

CH

MOLECULAR FORMULA

- Exact no. of atom present in molecule.
- C_6H_6

Calculation of EF & MF

$$MF = n \times EF$$

$$n = \frac{\text{Molecular mass}}{\text{Empirical mass}}$$

Stoichiometry and stoichiometric calculation

- The word "stoichiometry" comes from two Greek words:
  - "Stoicheion" means element
  - "Metron" means measure

- **Stoichiometry** is the study of how much of each substance (reactant and product) is involved in a chemical reaction.
- The numbers written in front of molecules in a balanced chemical equation are called stoichiometric coefficients. e.g $1CH_4 + \underline{2}O_2 \longrightarrow 1CO_2 + \underline{2}H_2O$.
- These numbers tell us how many molecules or moles of each substance are taking part in the reaction.
- Doing calculations based on chemical equations is called stoichiometric calculations.

e.g. For a following Reaction

$$1\,CH_{4(g)} + 2\,O_{2(g)} \longrightarrow 1\,CO_{2(g)} + 2\,H_2O_{(g)}$$

$12+4=16g$

We can say that,

In Stoichiometry and limiting Reagent reactions must be balanced.

in terms of moles.

- One mole of methane (CH_4) reacts with 2 moles of O_2 to produce 1 mole of CO_2 & 2 mole of H_2O.

In terms of molecule

1 molecule of methane reacts with 2 molecules of oxygen (O_2) to produce 1 molecule of CO_2 & 2 molecule of H_2O.

In terms of molar mass.

16g of methane reacts with 64g of oxygen (O_2) to produce 44g of CO_2 & 36g of H_2O.

In terms of Volume :→

22.4 L of CH_4 reacts with (2×22.4) L to produce 22.4 L of CO_2 & 44.8 L of H_2O.

LIMITING REAGENT OR LIMITING REACTANT

* Important

- The reactant which is present in the lesser amount and gets consumed after sometime i.e. which limits the amount of product formed is called the limiting reagent. or reactant.
- The reactant other than the limiting reagent, which is in somewhat excess is called the excess reagent.

e.g $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

16g 64g 44g 36g

As per this equation 16g of CH_4 is required to react with 64g of O_2. but if 15g of CH_4 is given then CH_4 will act as limiting reagent & O_2 will act as excess reagent.

EXAMPLE

1 atom 1 molecule 1 molecule.

In a reaction, $\boxed{1A + 1B_2 \longrightarrow AB_2}$ identify the limiting reagent if any in the following reaction mixtures.

Limiting Reagent

(i) 300 atoms of A + 200 molecules of B → B_2.

(ii) 2 moles of A + 3 moles of B → A

(iii) 100 atoms of A + 100 molecules of B → No L.R.

(iv) 5 moles of A + 2.5 moles of B_2 → B_2

(v) 2.5 moles of A + 5 moles of B_2 → A.

EXAMPLE

Calculate the amount of water (g) produced by the combustion of 16 g of methane.

Sol:→

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

Molar mass.

16g $2(32) = 64g$ $12+32 = 44g$ 36g

Given

16g —

16g of CH_4 produces 36g of H_2O.

EXAMPLE

How many moles of methane are required to produce 22 g $CO_2(g)$ after combustion?

Sol :→

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

Molar mass	16g	64g	44g	36g
Given	?		22g	

44g of CO_2 is produced by = 16g of CH_4.

1g of " " " " = $\frac{16g}{44g}$ of CH_4.

22g of " " " " = $\frac{\cancel{16}^{8}}{\cancel{44}} \times \cancel{22}$ of CH_4

= 8g of CH_4.

No. of moles of methane = $\frac{\text{Give mass}}{\text{Molar mass}}$

$= \frac{8g}{16g}$

$= 0.5$ mole

Q. 3g of H_2 react with 29g of O_2 to yield H_2O.

a) which is L.R.

b). Calculate the amount of H_2O produced.

c) Calculate the amount of excess Reagent.

Sol :→ a). Balanced Chemical equal equation

$$2H_2 + O_2 \longrightarrow 2H_2O$$

Molar mass: 4g, 32g, 36g

4g of H_2 is required for = 32g of O_2.

1g of " " " " = $\frac{32}{4}$ of O_2

3g of H_2 " " " = $\frac{32}{4} \times 3$ of of O_2 = 24g

(PYQ).

But we have 29g of O_2 i.e it is in excess.

hence H_2 is given is lesser amount i.e H_2 is limiting Reagent.

b) 4g of H_2 produces = 36g of H_2O.

1g of H_2 " = $\frac{36}{4}$ g of H_2O.

3g of H_2 " = $\frac{36}{4} \times 3$ g of H_2O

= 27g of H_2O.

c) Amount of excess Reagent.

= Given reactant − Required Reactant

= 29g − 24g

= 5g ✓

EXAMPLE

Dinitrogen and dihydrogen react with each other to produce ammonia according to the following chemical equation,

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3\,(g)$$

(i) Calculate the mass of ammonia produced if $2.00 \times 10^3 g$ dinitrogen reacts with $1.00 \times 10^3\ g$ of dihydrogen.

(ii) Will any of the two reactants remain unreacted?

(iii) If yes, which one and what would be its mass?

EXAMPLE

Chlorine is prepared in the laboratory by treating manganese dioxide (MnO_2) with aqueous hydrochloric acid according to the reaction,

$$4\ HCl\ (aq) + MnO_2\ (s) \longrightarrow 2H_2O\ (l) + MnCl_2\ (aq) + Cl_2(g)$$

How many gram of HCl reacts with 5.0 g of manganese dioxide?

QUESTION

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Lallu

Match the following terms with their correct descriptions:

Column – I		Column – II	
A.	Stoichiometry	1.	Smallest amount of reactant that limits the product
B.	Limiting Reagent	2.	Deals with calculation of moles in a reaction
C.	Balanced Equation	3.	Equation with equal number of atoms on both sides
D.	Mole	4.	SI unit for amount of substance

1. A–2, B–1, C–3, D–4
2. A–3, B–2, C–1, D–4
3. A–1, B–3, C–4, D–2
4. A–4, B–3, C–2, D–1

QUESTION

Reactant OR

What is a limiting reagent?

1. The reactant in excess (excess Reagent)
2. The reactant that is completely consumed
3. The product formed in lesser amount
4. None of the above

CHAPTER KHATAM.

1. Murra
2. Zeesham
3. Aleem.

Thank You